

Vemula Sai Kumar

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Profile Summary

Cybersecurity professional with ~4 years of experience in IT and security operations, specializing in SIEM (Splunk), threat detection, and incident response. Skilled in vulnerability management, penetration testing, and cloud security (AWS), with experience in automating workflows to improve detection efficiency and reduce alert fatigue.

Education

Yeshiva University, Katz School of Science and Health

New York, NY

Master of Science in Cybersecurity

May 2026

Overall GPA: 3.9

CMR College of Engineering and Technology

Hyderabad, India

Bachelor of Technology.

Sep 2021

Overall GPA: 3.9

Skills

- **Security Operations & Monitoring:** SIEM (Splunk), Security Monitoring, Log Analysis, Alert Investigation, Event Correlation, Incident Detection & Response, Basic Threat Hunting, MITRE ATT&CK
- **Network Security & Analysis:** Wireshark, Packet Analysis, TCP/IP, DNS, HTTP/HTTPS, VPNs, Firewalls, Network Traffic Inspection, IDS/IPS Concepts (Suricata, Zeek)
- **Cloud Security & Risk Management:** AWS (IAM, S3, EC2, Security Groups, RDS), Azure & GCP (basic), Cloud Misconfigurations, Risk Assessment, Threat Modeling, Compliance (HIPAA, PCI DSS, SOC 2, ISO 27001)
- **Vulnerability & Web Security:** Vulnerability Assessment (DVWA), OWASP Top 10 (SQL Injection, XSS, CSRF, Command Injection, Brute Force), Burp Suite, Kali Linux, Security Testing & Mitigation
- **Automation & Scripting:** Python (automation & scripting), FastAPI, REST APIs, WebSockets, Scikit-learn (basic ML), TensorFlow (basic), Bash
- **Tools & Documentation:** Microsoft Excel, Word, PowerPoint, SharePoint, Office 365, Adobe Acrobat
- **Soft Skills:** Project Management, Critical Thinking, Analytical Skills, Communication, Attention to Detail, Initiative-Driven, Curiosity, Resilience

Work Experience

Cybersecurity Analyst L1 | Deloitte - Pune, India

Mar 2023 – Apr 2025

- Executed red team simulations on enterprise banking infrastructure using Metasploit and Burp Suite, identifying exploitable attack paths and strengthening **penetration testing and threat detection capabilities**.
- Led vulnerability management efforts using Nessus and Qualys, combining automated scanning with manual validation to support **risk prioritization and remediation**, reducing critical exposures by **18%**.
- Designed and optimized **SIEM detection rules in Splunk**, improving **security monitoring, alert correlation, and incident detection** for privilege escalation and persistence techniques.
- Performed **network traffic analysis and threat hunting** using Wireshark and Zeek, identifying indicators of compromise (IoCs) such as beaconing activity and lateral movement.
- Conducted **cloud security assessments in AWS**, identifying IAM misconfigurations and privilege escalation risks, improving **cloud security posture management (CSPM)** and reducing risks by **35%**.
- Led **purple team exercises**, collaborating across SOC and red team functions to enhance **incident response workflows, detection engineering, and SOC efficiency**, improving response time by **28%**.
- Developed automated workflows within **SIEM and SOAR platforms**, enabling faster **alert triage, incident response automation, and SOC operations optimization**.

Cybersecurity Engineer | Phoenix Business Consulting Pvt Ltd

Aug 2022 – Feb 2023

- Led **Red Team adversary simulations and penetration testing** using Metasploit and Burp Suite, identifying critical vulnerabilities aligned with **OWASP Top 10**, reducing exploitable attack surface by **~20%**.
- Engineered **Blue Team detection capabilities in SIEM (Splunk)**, improving **threat detection accuracy and alert correlation**, resulting in a **~30% improvement in detection coverage**.
- Executed **vulnerability management lifecycle** using Nessus and Qualys, performing manual validation and prioritization of risks, reducing high-risk vulnerabilities by **~25%**.
- Conducted **threat hunting and network traffic analysis** using Wireshark and Zeek, identifying **IoCs** such as lateral movement and C2 beaconing, strengthening proactive threat detection.
- Built **Python-based automation workflows** for log analysis and alert triage, reducing **SOC alert fatigue by ~35%** and improving Tier 1/Tier 2 operational efficiency.

IT Support Technician | Siddhartha Heavy Equipment Limited

Jun 2021 – Jul 2022

- Installed, configured, and maintained **Windows systems and applications**, ensuring system performance and reliability.
- Troubleshoot **network connectivity issues (TCP/IP, DNS, HTTP/S, VPN, Wi-Fi)** to support stable operations across users.
- Managed **user onboarding and offboarding**, including system setup and secure access provisioning aligned with **access control policies**.
- Documented **incidents, service requests, and troubleshooting steps** in IT ticketing systems, improving resolution efficiency.
- Supported **endpoint security** by applying system patches, antivirus updates, and monitoring for potential security threats.
- Followed **IT security policies, incident response procedures, and cybersecurity best practices**, escalating suspicious activity when required.

Projects

AI-Powered Cloud Security Posture & Deception Platform | *Python, AWS, Azure, TensorFlow, Scikit-learn, Terraform*

- Developed an **AI-driven cloud security platform** for **cloud security posture management (CSPM)**, detecting misconfigurations across AWS services (IAM, S3, EC2).
- Performed **cloud risk assessment and threat modeling**, mapping threats to **MITRE ATT&CK and CIS Benchmarks** for compliance and security analysis.
- Implemented **deception technology (honeypots)** and **real-time alerting**, improving **threat detection, incident response, and MTTD/MTTR**.

Web Application Vulnerability Testing (DVWA) | *Kali Linux, Burp Suite, DVWA*

- Conducted vulnerability assessment and penetration testing (VAPT) on OWASP Top 10 vulnerabilities (SQL Injection, XSS, CSRF, Brute Force).
- Identified and exploited vulnerabilities, applying **secure coding practices, input validation, and access control mechanisms**.

Smart SIEM Risk Engine | *Python, FastAPI, WebSockets, SQLite, Scikit-learn, MITRE ATT&CK*

- Built an **automated SIEM solution** for **security monitoring, log analysis, and threat detection**, reducing **SOC alert fatigue**.
- Developed a **machine learning-based risk scoring engine** (Random Forest) for **threat classification and alert prioritization**.
- Implemented **incident response automation and detection engineering**, with real-time dashboards, GeoIP visualization, and alert notifications. GitHub <https://github.com/svemula17/smart-siem-risk-engine>

Certifications

- CompTIA Security + CompTIA
- CySA+ (in progress)
- Google Cybersecurity Professional Certificate (Coursera)
- Certified in Cybersecurity Specialization — (ISC)² / Coursera